Recursion

Programming · Level 4 · Lesson 2



In this lesson you will

- Write recursive functions with a base case and a recursive case.
- Follow the call stack and know its limits.
- Speed up repeated work with memoisation.

Key ideas

- Base case first; each call on a smaller problem.
- The stack has limits: about 150 calls on the phone.
- Repeated work? Remember answers in a dictionary.

Activity: The human call stack

1. Warm-up (3 min): Five people in a line count who is behind them.
2. Act (7 min): Act out fact(4): each person waits for the next.
3. Count (7 min): Draw fib(5)’s call tree and count the repeats.
4. Fix (3 min): Add a “memory” sheet; how many calls now?

A paper call tree makes repeated work visible before any code runs.

Let's discuss

1. When is a loop better than recursion?
2. Why does fib repeat so much work?
3. What problems around you are made of smaller copies of themselves?

Exit ticket

- Write the base case and the recursive case of a function that adds up a list.